Exploratory Data Analysys

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This code uses static data as its starting point rather than running a lengthy API call. For production, to access realtime current data, this would be replaced by a call to the API using the reticulate functions that call Python code in R

```
RawData <- readRDS(here::here("Data","PreEDA_DataFrame.rds"))
names(RawData) <- make.unique(names(RawData), sep = ".")
FieldActions <- read_excel(here::here("Data", "FieldActions.xlsx"))
CleanedData <- RawData # make a copy to preserve original
IncomeData <- read_excel(here::here("Data", "Income Group Data.xlsx")) # Import income data
```

```
# This code iterates through every field in the raw data, taking the cleanup action on
# it that is specified in the FieldActions data frame.  The net result is that a bunch of
# duplicated and/or unneeded fields are deleted, fields with uninterpretable (coded) column
# names tied to their source are renamed to be descriptive, and the Year column is forced
# to an integer

for (fieldcounter in names(CleanedData)) {
  action_row <- FieldActions[FieldActions$FieldName == fieldcounter, ]
  if (nrow(action_row) == 0) {
    next  # No action specified for this field
  }
  WhatToDo <- action_row$Action
  if (WhatToDo == "Delete") {
    CleanedData[[fieldcounter]] <- NULL

  } else if (WhatToDo == "Rename") {
    names(CleanedData)[names(CleanedData) == fieldcounter] <- action_row$RenamedTo

  } else if (WhatToDo == "Make Integer") {
```

```r
    CleanedData[[fieldcounter]] <- as.integer(CleanedData[[fieldcounter]])
  }
}
# Now, pull in the Income Group Data, and join it to the raw data by CountryCode
CleanedData <- left_join(CleanedData, IncomeData, by = c("CountryCode" = "Country Code"))

# Now for initial EDA purposes, I am only going to consider the two wellbeing indexes
# which are tagged as my PrimaryResponse fields.  So I want to drop the other more granular
# Response fields

# Select field names to keep
# First, all the renamed fields that weren't dropped and aren't a Response
PrimaryFields <- FieldActions %>%
  filter(Type %in% c("PrimaryResponse","Predictor","Identifier"),
         is.na(Action) | Action != "Delete") %>%
  pull(RenamedTo)
# Now add the Income Group field (since that was joined from a different data set)
PrimaryFields <- c(PrimaryFields, "Income Group")
# Subset the cleaned dataset using that list of fields
FirstCutData <- CleanedData %>%
  select(all_of(PrimaryFields))
write_xlsx(FirstCutData, path = here::here("Data","FirstCutData.xlsx"))
```

Table 1: Summary Statistics (Variables 1 to 8)

CountryCode	CountryName	year	HDI_Index	IHDI_Index	GiniCoeff	FoodIndex	CorruptionScore
Length:5974	Length:5974	Min. :1995	Min. :0.2380	Min. :0.1340	Min. : 4.819	Min. : 17.00	Min. :-1.96956
Class :character	Class :character	1st Qu.:2002	1st Qu.:0.5700	1st Qu.:0.3990	1st Qu.:11.060	1st Qu.: 76.58	1st Qu.:-0.82041
Mode :character	Mode :character	Median :2009	Median :0.7110	Median :0.5930	Median :19.183	Median : 94.52	Median :-0.29892
NA	NA	Mean :2009	Mean :0.6899	Mean :0.5783	Mean :21.317	Mean : 91.33	Mean :-0.05974
NA	NA	3rd Qu.:2016	3rd Qu.:0.8140	3rd Qu.:0.7500	3rd Qu.:31.878	3rd Qu.:102.80	3rd Qu.: 0.59246
NA	NA	Max. :2023	Max. :0.9720	Max. :0.9230	Max. :51.891	Max. :578.71	Max. : 2.45912
NA	NA	NA	NA's :422	NA's :3677	NA's :3670	NA's :720	NA's :1204

Table 2: Summary Statistics (Variables 9 to 16)

ElectricAccess	PopDensity	PopInSlums	WaterStress	BankingAccess	RnDInvest	GovtEffectiveness	ShippingIndex
Min. : 0.80	Min. : 1.438	Min. : 0.00	Min. : 0.0258	Min. : 0.40	Min. :0.0054	Min. :-2.44023	Min. : 0.6032
1st Qu.: 63.20	1st Qu.: 30.875	1st Qu.: 1.55	1st Qu.: 3.7148	1st Qu.: 31.07	1st Qu.:0.2383	1st Qu.:-0.78651	1st Qu.: 7.5392
Median : 98.10	Median : 77.922	Median :21.18	Median : 11.1751	Median : 54.05	Median :0.6029	Median :-0.20432	Median : 15.1677
Mean : 79.23	Mean : 298.200	Mean :28.73	Mean : 65.0840	Mean : 57.04	Mean :0.9675	Mean :-0.06367	Mean : 25.1422
3rd Qu.:100.00	3rd Qu.: 176.610	3rd Qu.:52.80	3rd Qu.: 39.6974	3rd Qu.: 86.76	3rd Qu.:1.3904	3rd Qu.: 0.59433	3rd Qu.: 34.1564
Max. :100.00	Max. :18822.891	Max. :99.81	Max. :3850.5000	Max. :100.00	Max. :6.0192	Max. : 2.46966	Max. :171.1775
NA's :627	NA's :583	NA's :4011	NA's :1505	NA's :5412	NA's :3666	NA's :1223	NA's :3592

Table 3: Summary Statistics (Variables 17 to 24)

FixedIntSubs	InternetUsersPct	PoliticalStability	RuleOfLaw	RqdEduYears	GovtEduSpendPctGDP	MortalityFromDirtyness	UHCServiceCoverage
Min. : 0.0000	Min. : 0.00	Min. :-3.31295	Min. :-2.59088	Min. : 4.00	Min. : 0.000	Min. : 0.40	Min. :11.00
1st Qu.: 0.2673	1st Qu.: 2.81	1st Qu.:-0.69566	1st Qu.:-0.82174	1st Qu.: 8.00	1st Qu.: 3.139	1st Qu.: 2.95	1st Qu.:44.00
Median : 3.8683	Median : 21.00	Median : 0.02612	Median :-0.20532	Median : 9.00	Median : 4.232	Median : 6.10	Median :63.00
Mean :10.5358	Mean : 32.58	Mean :-0.06279	Mean :-0.06373	Mean : 9.42	Mean : 4.418	Mean : 16.96	Mean :59.39
3rd Qu.:18.4017	3rd Qu.: 61.40	3rd Qu.: 0.79477	3rd Qu.: 0.69307	3rd Qu.:11.00	3rd Qu.: 5.428	3rd Qu.: 25.55	3rd Qu.:75.00
Max. :62.2147	Max. :100.00	Max. : 1.75868	Max. : 2.12476	Max. :17.00	Max. :15.863	Max. :108.10	Max. :91.00
NA's :2041	NA's :675	NA's :1160	NA's :1114	NA's :1432	NA's :2288	NA's :5791	NA's :4630

Table 4: Summary Statistics (Variables 25 to 32)

HealthSpendPerCapita	GovtHealthSpendPerCapita	MigrantsPct	FoodInsecurityPct	RuralPopulGrowth	VoiceAccountability	Homicides	Income Group
Min. : 4.176	Min. : 0.1489	Min. : 0.036	Min. : 2.00	Min. :-235.9259	Min. :-2.31340	Min. : 0.000	Length:5974
1st Qu.: 63.631	1st Qu.: 21.9384	1st Qu.: 1.212	1st Qu.: 8.90	1st Qu.: -0.5789	1st Qu.:-0.88313	1st Qu.: 1.228	Class :character
Median : 261.295	Median : 128.2416	Median : 3.663	Median :22.30	Median : 0.4574	Median :-0.00619	Median : 2.861	Mode :character
Mean : 959.978	Mean : 648.7002	Mean : 8.791	Mean :29.09	Mean : 0.3933	Mean :-0.04756	Mean : 7.939	NA
3rd Qu.: 890.483	3rd Qu.: 557.3557	3rd Qu.:10.632	3rd Qu.:45.30	3rd Qu.: 1.6167	3rd Qu.: 0.85004	3rd Qu.: 8.879	NA
Max. :12434.434	Max. :7871.2450	Max. :88.404	Max. :89.20	Max. : 15.0306	Max. : 1.80099	Max. :138.774	NA
NA's :1617	NA's :1630	NA's :5009	NA's :4953	NA's :458	NA's :1112	NA's :2755	NA

Table 5: Data summary

Name	FirstCutData
Number of rows	5974
Number of columns	32
Column type frequency:	
character	3
numeric	29
Group variables	None

Variable type: character

skim_variable	n_missing	complete_rate	min	max	empty	n_unique	whitespace
CountryCode	0	1.00	3	9	0	206	0
CountryName	319	0.95	4	38	0	195	0
Income Group	377	0.94	10	19	0	4	0

Variable type: numeric

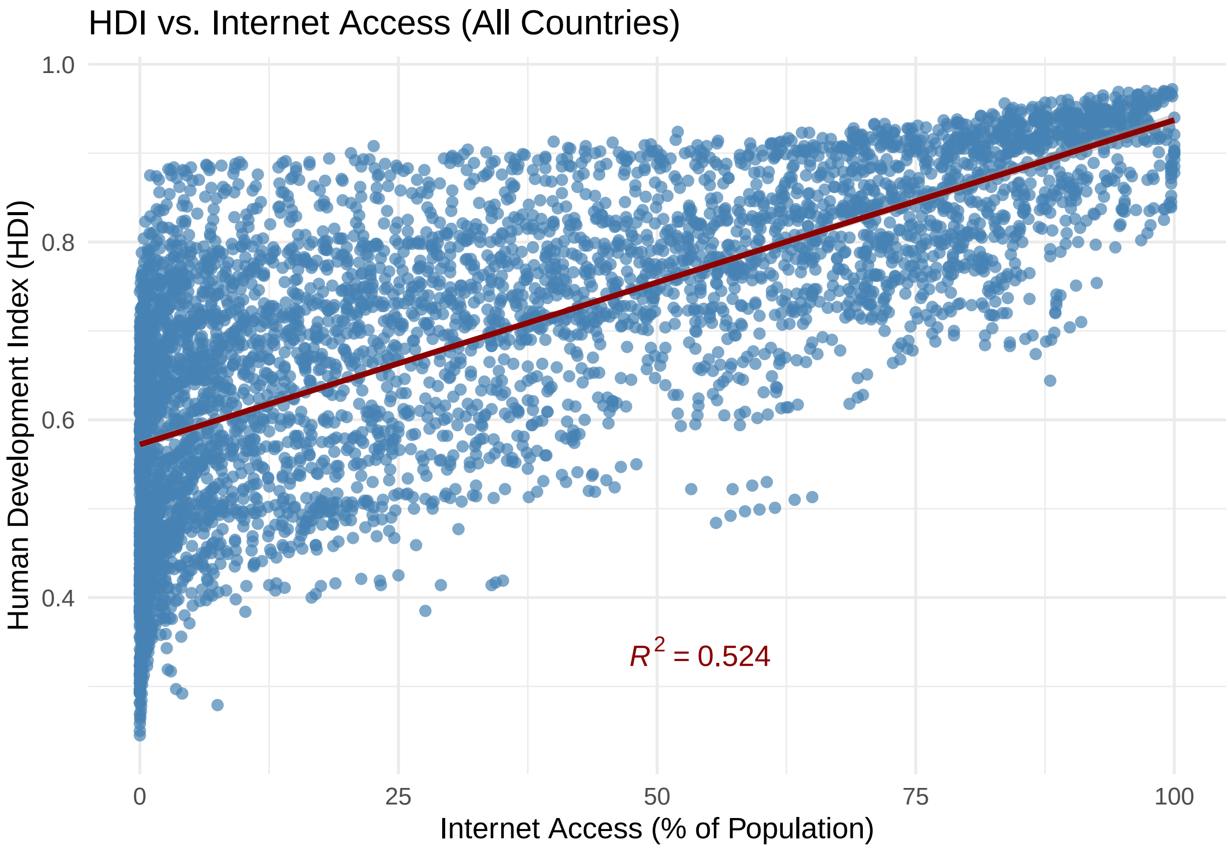
skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100	hist
year	0	1.00	2009.00	8.37	1995.00	2002.00	2009.00	2016.00	2023.00	
HDI_Index	422	0.93	0.69	0.16	0.24	0.57	0.71	0.81	0.97	
IHDI_Index	3677	0.38	0.58	0.20	0.13	0.40	0.59	0.75	0.92	
GiniCoeff	3670	0.39	21.32	11.35	4.82	11.06	19.18	31.88	51.89	
FoodIndex	720	0.88	91.33	28.06	17.00	76.57	94.52	102.80	578.71	
CorruptionScore	1204	0.80	-0.06	1.00	-1.97	-0.82	-0.30	0.59	2.46	
ElectricAccess	627	0.90	79.23	30.10	0.80	63.20	98.10	100.00	100.00	
PopDensity	583	0.90	298.20	1389.80	1.44	30.87	77.92	176.61	18822.89	
PopInSlums	4011	0.33	28.73	27.32	0.00	1.55	21.18	52.80	99.81	
WaterStress	1505	0.75	65.08	266.68	0.03	3.71	11.18	39.70	3850.50	
BankingAccess	5412	0.09	57.04	30.13	0.40	31.06	54.05	86.76	100.00	
RnDInvest	3666	0.39	0.97	0.98	0.01	0.24	0.60	1.39	6.02	
GovtEffectiveness	1223	0.80	-0.06	1.00	-2.44	-0.79	-0.20	0.59	2.47	
ShippingIndex	3592	0.40	25.14	25.22	0.60	7.54	15.17	34.16	171.18	

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100	hist
FixedIntSubs	2041	0.66	10.54	13.07	0.00	0.27	3.87	18.40	62.21	
InternetUsersPct	675	0.89	32.58	32.11	0.00	2.81	21.00	61.40	100.00	
PoliticalStability	1160	0.81	-0.06	1.00	-3.31	-0.70	0.03	0.79	1.76	
RuleOfLaw	1114	0.81	-0.06	0.99	-2.59	-0.82	-0.21	0.69	2.12	
RqdEduYears	1432	0.76	9.42	2.25	4.00	8.00	9.00	11.00	17.00	
GovtEduSpendPctGDP	2288	0.62	4.42	1.89	0.00	3.14	4.23	5.43	15.86	
MortalityFromDirtyness	5791	0.03	16.96	21.65	0.40	2.95	6.10	25.55	108.10	
UHCServiceCoverage	4630	0.22	59.39	19.02	11.00	44.00	63.00	75.00	91.00	
HealthSpendPerCapita	1617	0.73	959.98	1689.18	4.18	63.63	261.29	890.48	12434.43	
GovtHealthSpendPerCapita	1630	0.73	648.70	1235.08	0.15	21.94	128.24	557.36	7871.24	
MigrantsPct	5009	0.16	8.79	13.58	0.04	1.21	3.66	10.63	88.40	
FoodInsecurityPct	4953	0.17	29.09	23.01	2.00	8.90	22.30	45.30	89.20	
RuralPopulGrowth	458	0.92	0.39	3.70	-235.93	-0.58	0.46	1.62	15.03	
VoiceAccountability	1112	0.81	-0.05	1.00	-2.31	-0.88	-0.01	0.85	1.80	
Homicides	2755	0.54	7.94	12.47	0.00	1.23	2.86	8.88	138.77	

```r
# Filter complete cases again (just in case)
ggplot(FirstCutData, aes(x = InternetUsersPct, y = HDI_Index)) +
  geom_point(alpha = 0.7, color = "steelblue") +
  geom_smooth(method = "lm", se = TRUE, color = "darkred") +
  stat_poly_eq(
    aes(label = after_stat(rr.label)),
    formula = y ~ x,
    parse = TRUE,
    output.type = "expression",
    rr.digits = 3,
    label.x = 0.6,           # X-position (adjust if needed)
    label.y = 0.15,          # Y-position (bottom-ish)
    color = "darkred",       # Match regression line
    na.rm = TRUE
  ) +
  labs(
    title = "HDI vs. Internet Access (All Countries)",
    x = "Internet Access (% of Population)",
    y = "Human Development Index (HDI)"
  ) +
```

```r
  theme_minimal()
```

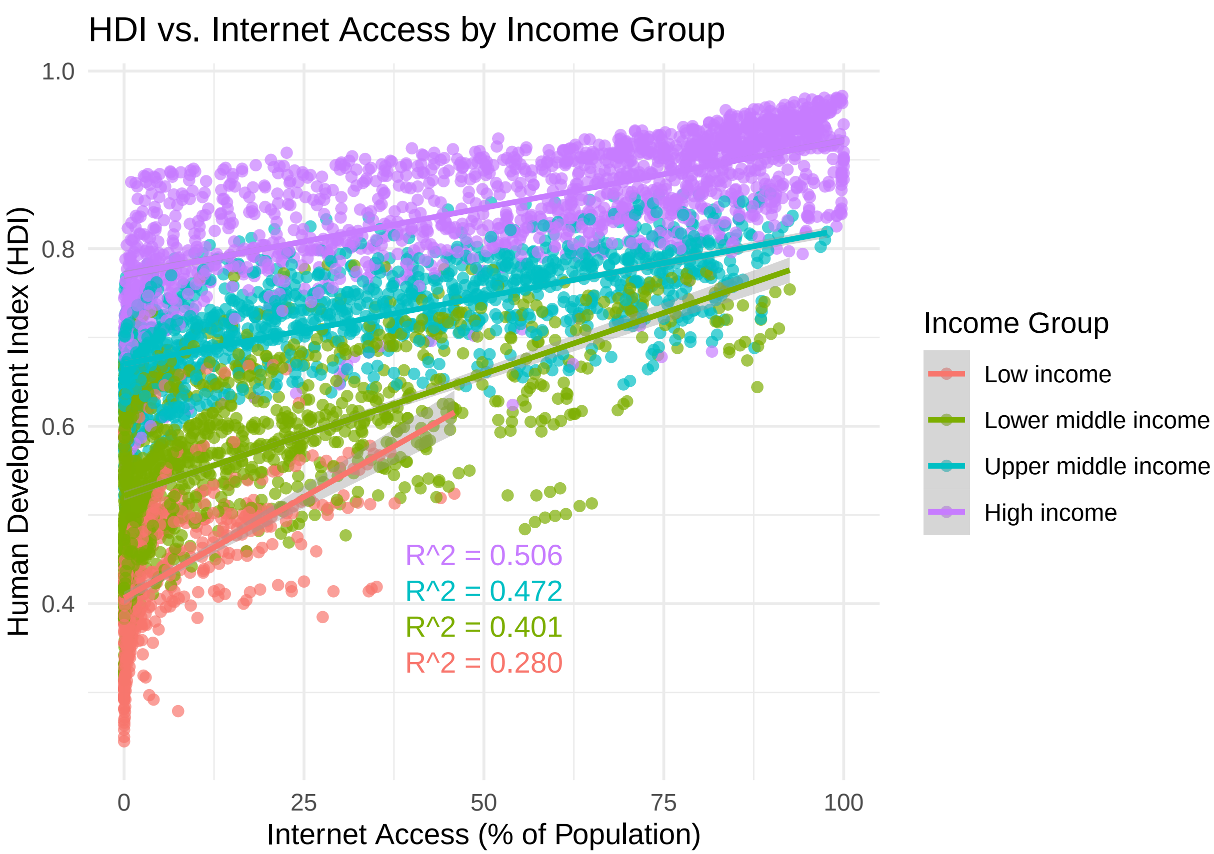
```
## `geom_smooth()` using formula = 'y ~ x'

## Warning: Removed 978 rows containing non-finite outside the scale range
## (`stat_smooth()`).

## Warning: Removed 978 rows containing missing values or values outside the scale range
## (`geom_point()`).
```




```r
FirstCutDataFiltered <- FirstCutData[!is.na(FirstCutData$`Income Group`), ]
FirstCutDataFiltered$`Income Group` <- factor(
  FirstCutDataFiltered$`Income Group`,
```

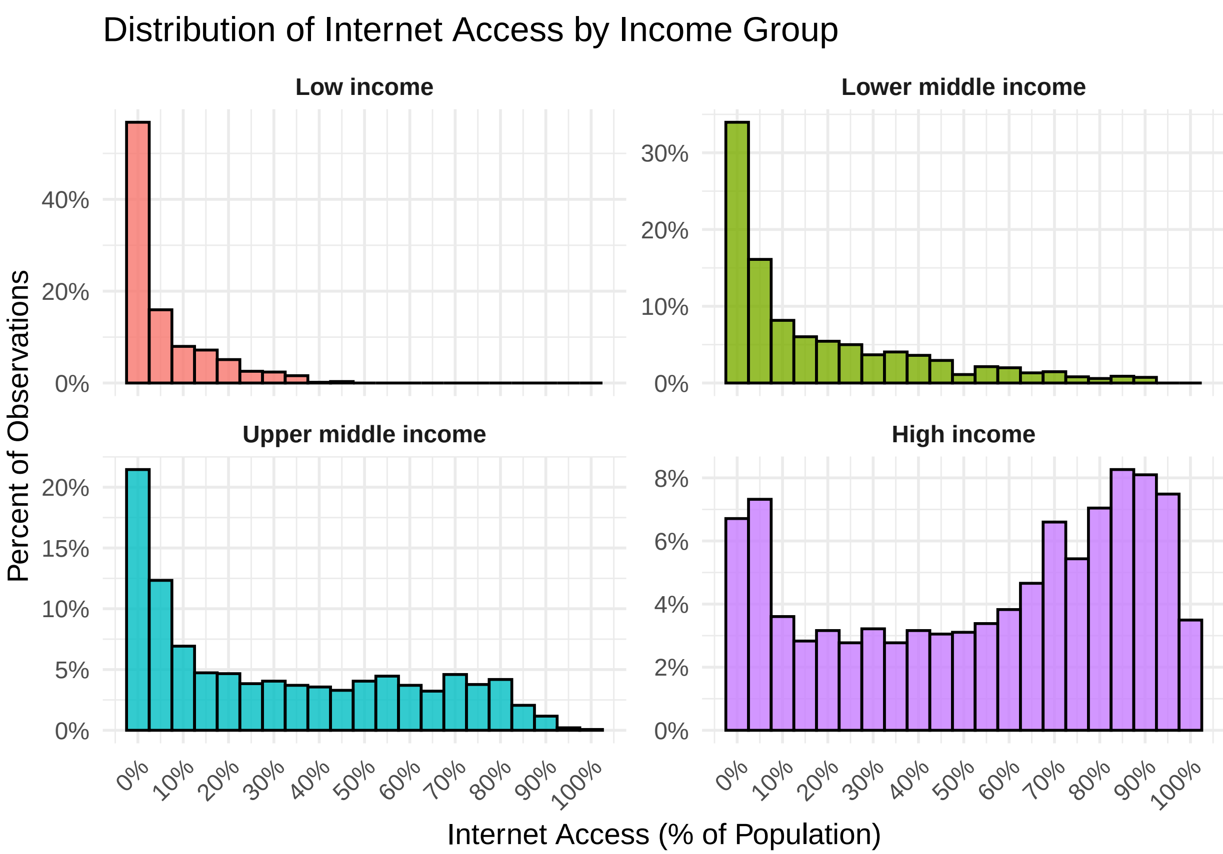
```r
  levels = c("Low income", "Lower middle income", "Upper middle income", "High income")
)
ggplot(FirstCutDataFiltered, aes(x = InternetUsersPct, y = HDI_Index, color = `Income Group`)) +
  geom_point(alpha = 0.7) +
  geom_smooth(method = "lm", se = TRUE) +
  stat_poly_eq(
    aes(
      label = after_stat(rr.label),
      group = `Income Group`,
      color = `Income Group`
    ),
    formula = y ~ x,
    parse = FALSE,
    output.type = "text",
    rr.digits = 3,
    label.x = 0.6,
    label.y = c(0.15, 0.2, 0.25, 0.3),  # Stack in bottom right
    na.rm = TRUE
  ) +
  labs(
    title = "HDI vs. Internet Access by Income Group",
    x = "Internet Access (% of Population)",
    y = "Human Development Index (HDI)",
    color = "Income Group"
  ) +
  theme_minimal()
```

```
## `geom_smooth()` using formula = 'y ~ x'

## Warning: Removed 646 rows containing non-finite outside the scale range
## (`stat_smooth()`).

## Warning: Removed 646 rows containing missing values or values outside the scale range
## (`geom_point()`).
```



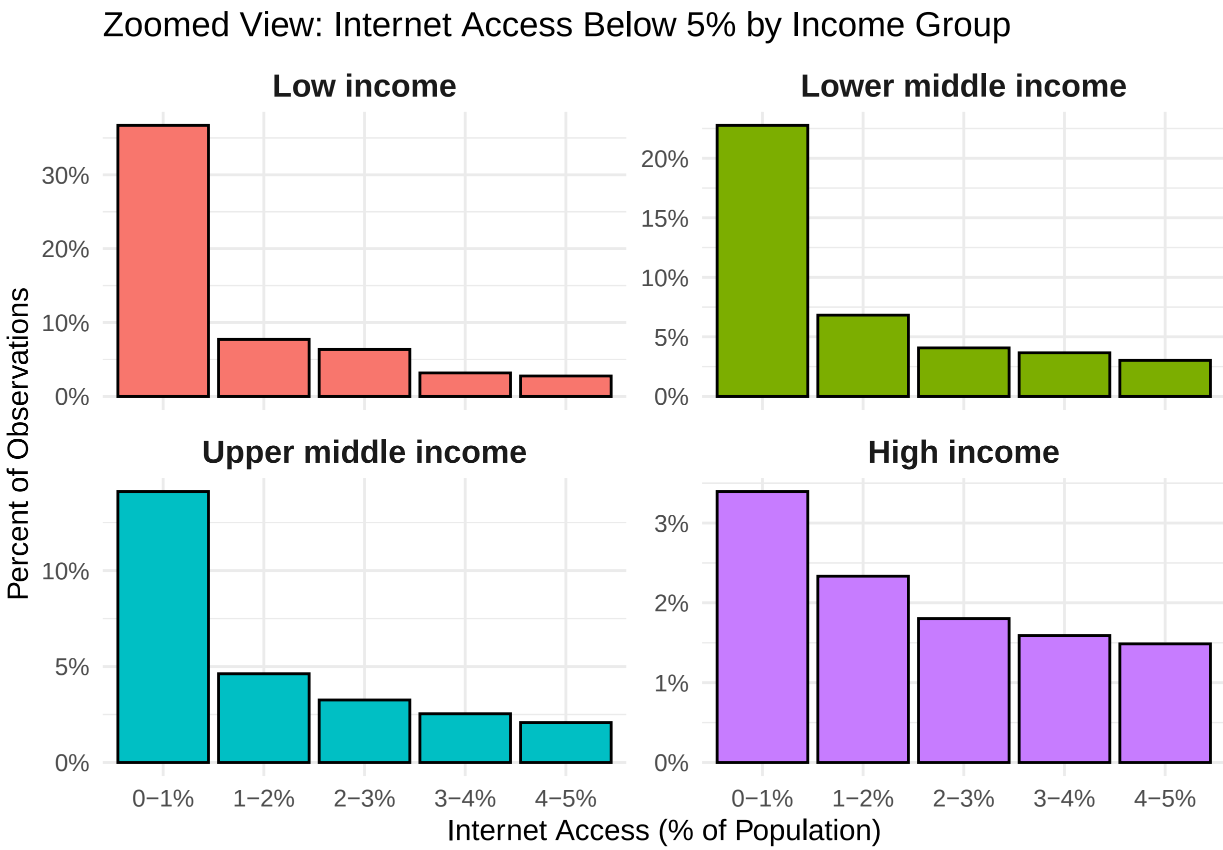

```
FirstCutDataFiltered <- FirstCutData[!is.na(FirstCutData$`Income Group`), ]

FirstCutDataFiltered$`Income Group` <- factor(
  FirstCutDataFiltered$`Income Group`,
  levels = c("Low income", "Lower middle income", "Upper middle income", "High income")
)

FirstCutDataFilteredClean <- FirstCutDataFiltered %>%
  filter(!is.na(InternetUsersPct))


ggplot(FirstCutDataFilteredClean, aes(x = InternetUsersPct, fill = `Income Group`)) +
```

```r
geom_histogram(
  aes(y = after_stat(count / tapply(count, PANEL, sum)[as.integer(PANEL)])),
  binwidth = 5,
  color = "black",
  alpha = 0.8
) +
facet_wrap(~ `Income Group`, scales = "free_y") +
scale_y_continuous(labels = scales::percent_format(accuracy = 1)) +
scale_x_continuous(
  breaks = seq(0, 100, by = 10),
  labels = function(x) paste0(x, "%")
) +
labs(
  title = "Distribution of Internet Access by Income Group",
  x = "Internet Access (% of Population)",
  y = "Percent of Observations"
) +
theme_minimal() +
theme(
  strip.text = element_text(face = "bold"),
  axis.text.x = element_text(angle = 45, hjust = 1),
  legend.position = "none"
)
```




```r
# Step 1: Filter to only rows with Internet Access under 5%
# Bin and summarize within the 0%-5% range
# Bin and summarize within the 0%-5% range
zoom_data <- FirstCutDataFiltered %>%
  filter(InternetUsersPct >= 0, InternetUsersPct < 5) %>%
  mutate(Bin = cut(
    InternetUsersPct,
    breaks = seq(0, 5, by = 1),
    include.lowest = TRUE,
    right = FALSE,
    labels = c("0-1%", "1-2%", "2-3%", "3-4%", "4-5%")
  ))
```

```r
# Group totals for each Income Group (full data)
group_totals <- FirstCutDataFiltered %>%
  count(`Income Group`, name = "GroupTotal")

# Bin counts and percentages
zoom_summary <- zoom_data %>%
  count(`Income Group`, Bin, name = "BinCount") %>%
  left_join(group_totals, by = "Income Group") %>%
  mutate(Percent = BinCount / GroupTotal)

# Plot
ggplot(zoom_summary, aes(x = Bin, y = Percent, fill = `Income Group`)) +
  geom_col(color = "black", width = 0.9) +
  facet_wrap(~ `Income Group`, scales = "free_y") +
  scale_y_continuous(labels = scales::percent_format(accuracy = 1)) +
  labs(
    title = "Zoomed View: Internet Access Below 5% by Income Group",
    x = "Internet Access (% of Population)",
    y = "Percent of Observations"
  ) +
  theme_minimal() +
  theme(
    strip.text = element_text(face = "bold", size = 12),
    axis.text.x = element_text(angle = 0, hjust = 0.5),
    legend.position = "none"
  )
```

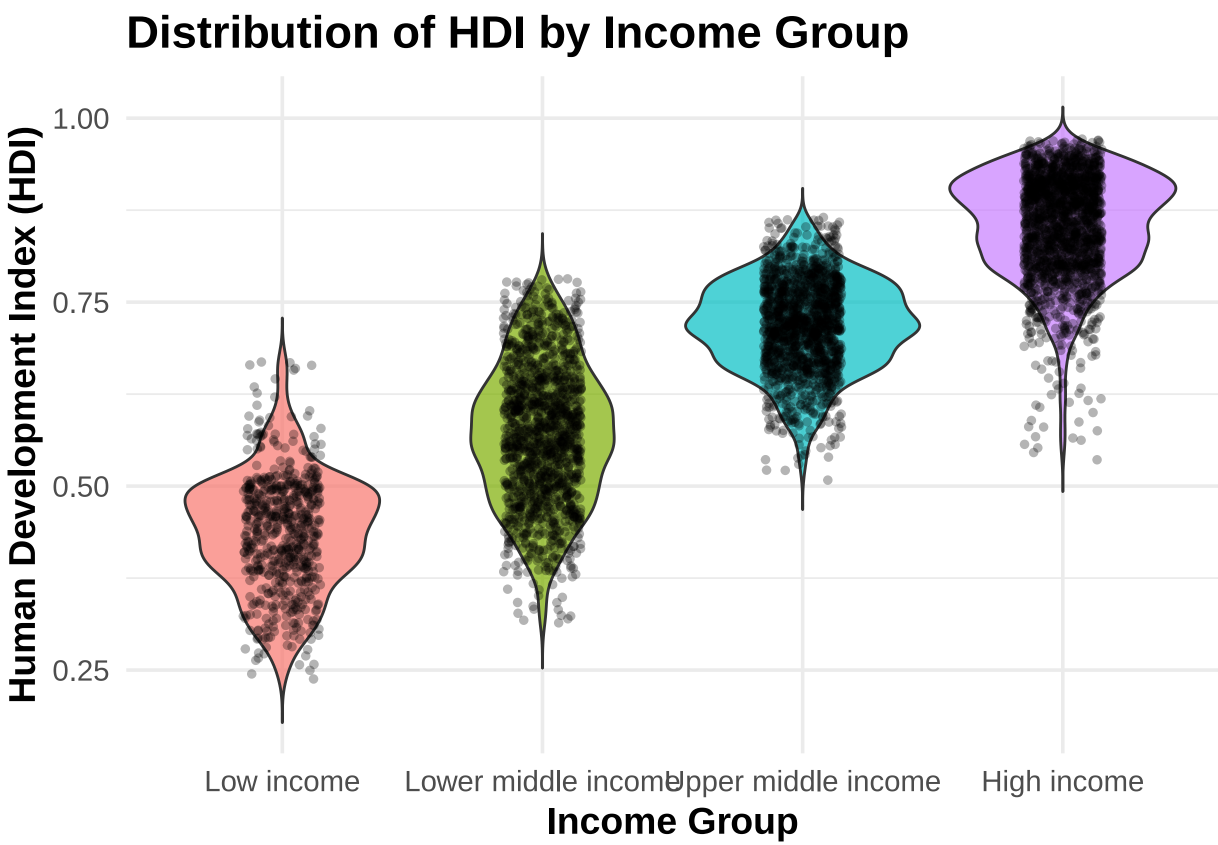



```r
ggplot(FirstCutDataFiltered, aes(x = `Income Group`, y = HDI_Index, fill = `Income Group`)) +
  geom_violin(trim = FALSE, alpha = 0.7) +
  geom_jitter(width = 0.15, alpha = 0.3, color = "black", size = 1) +
  labs(
    title = "Distribution of HDI by Income Group",
    x = "Income Group",
    y = "Human Development Index (HDI)"
  ) +
  theme_minimal(base_size = 14) +
  theme(
    legend.position = "none",
    plot.title = element_text(face = "bold"),
```

```
    axis.title.x = element_text(face = "bold"),
    axis.title.y = element_text(face = "bold")
  )
```

```
## Warning: Removed 417 rows containing non-finite outside the scale range
## (`stat_ydensity()`).
```

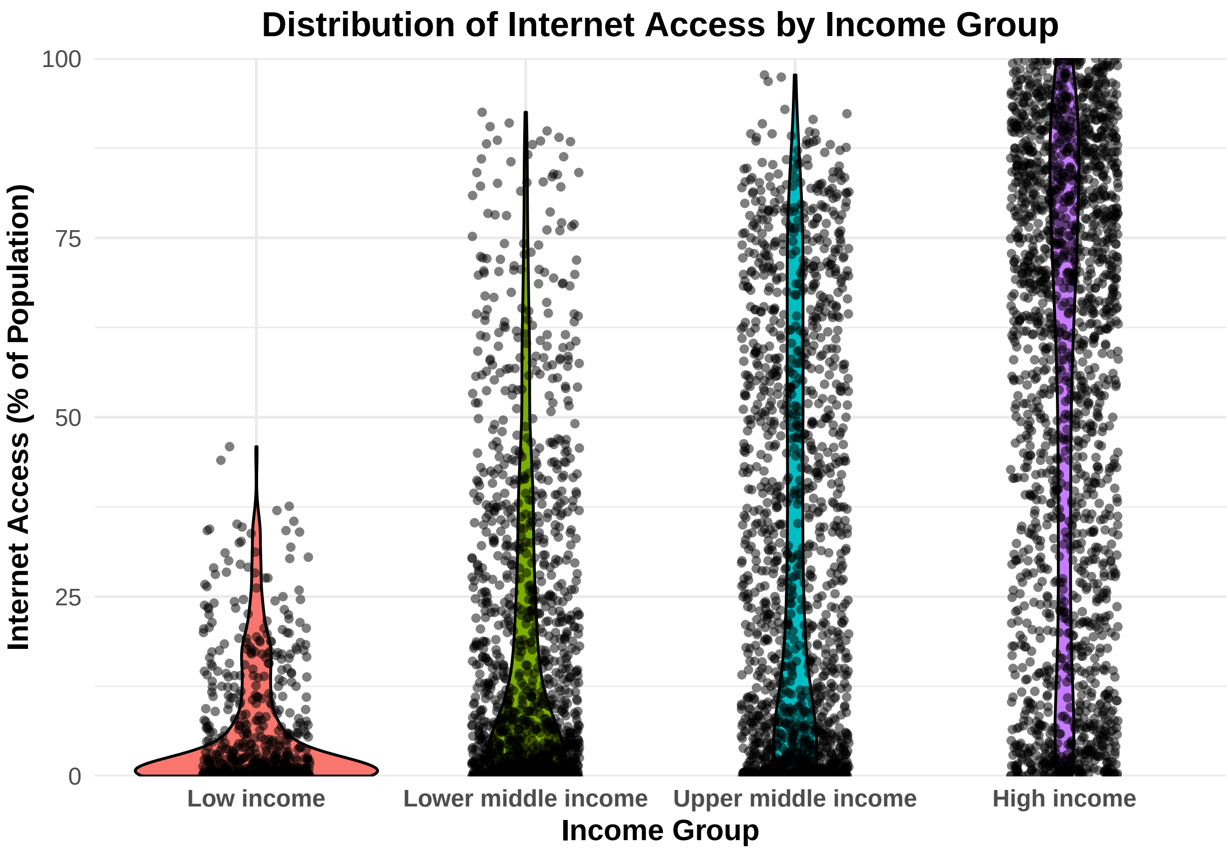
```
## Warning: Removed 417 rows containing missing values or values outside the scale range
## (`geom_point()`).
```




```
ggplot(FirstCutDataFiltered, aes(x = `Income Group`, y = InternetUsersPct, fill = `Income Group`)) +
  geom_violin(trim = TRUE, color = "black", adjust = 1.2) +  # trim removes tails beyond the data
  geom_jitter(width = 0.2, size = 1, alpha = 0.5, color = "black") +
```

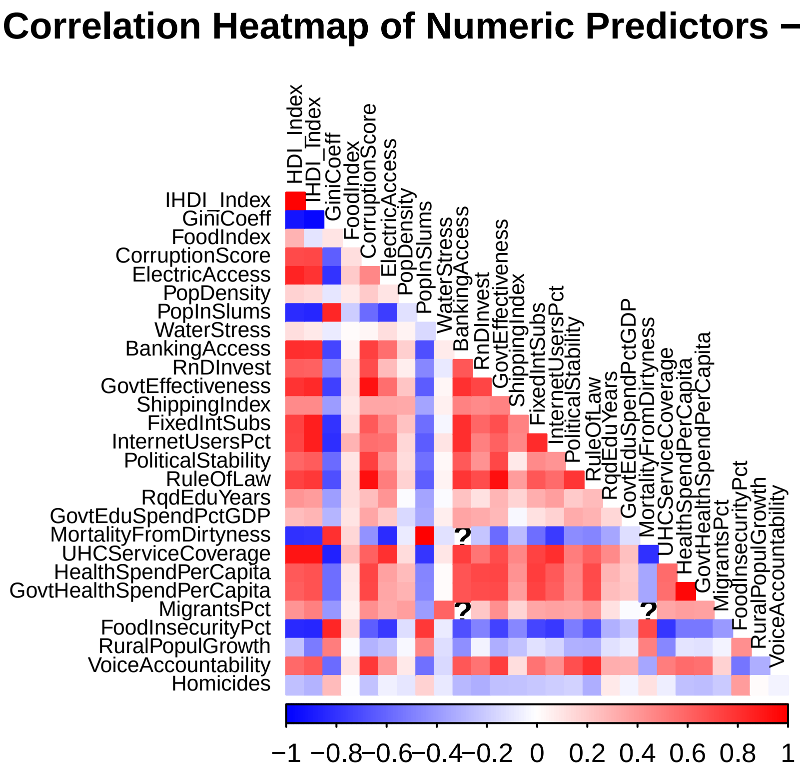
```r
  scale_y_continuous(limits = c(0, 100), expand = c(0, 0)) +  # Hard clip y-axis
  labs(
    title = "Distribution of Internet Access by Income Group",
    y = "Internet Access (% of Population)",
    x = "Income Group"
  ) +
  theme_minimal() +
  theme(
    plot.title = element_text(hjust = 0.5, face = "bold"),
    axis.title = element_text(face = "bold"),
    axis.text.x = element_text(face = "bold"),
    legend.position = "none"
  ) +
  scale_fill_manual(values = c(
    "Low income" = "#F8766D",
    "Lower middle income" = "#7CAE00",
    "Upper middle income" = "#00BFC4",
    "High income" = "#C77CFF"
  ))
```

```
## Warning: Removed 348 rows containing non-finite outside the scale range
## (`stat_ydensity()`).

## Warning: Removed 370 rows containing missing values or values outside the scale range
## (`geom_point()`).
```



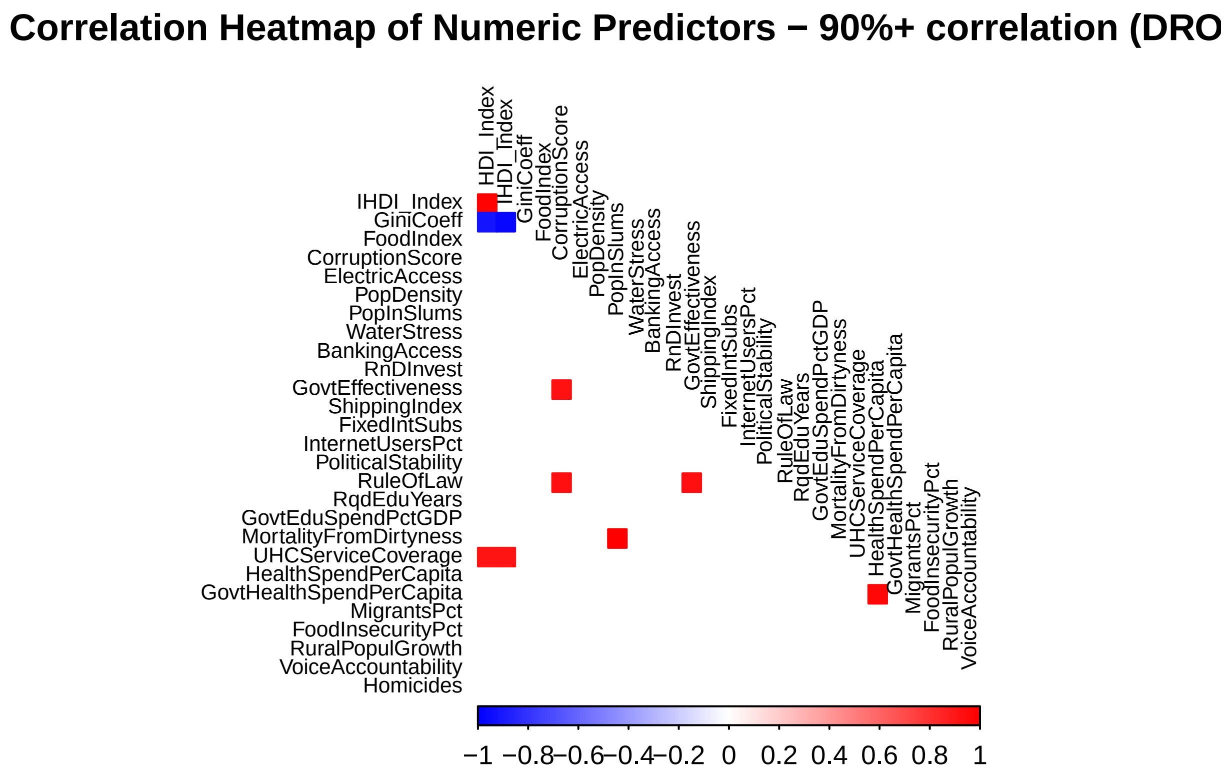

```
# Step 1: Select only numeric columns and drop unwanted ones
numeric_vars <- FirstCutData %>%
  select(where(is.numeric)) %>%
  select(-year)  # Drop the 'Year' column

# Step 2: Compute correlation matrix using pairwise complete observations
cor_matrix <- cor(numeric_vars, use = "pairwise.complete.obs", method = "pearson")
cor_matrix_90 <- ifelse(abs(cor_matrix) >= 0.9, cor_matrix, NA)
cor_matrix_70_90 <- ifelse(abs(cor_matrix) >= 0.7 & abs(cor_matrix) < 0.9, cor_matrix, NA)


# Step 3: Plot correlation heatmap
```

```r
corrplot(cor_matrix,
         method = "color",
         type = "lower",
         tl.col = "black",
         tl.cex = 0.7,
         col = colorRampPalette(c("blue", "white", "red"))(200),
         diag = FALSE)
title("Correlation Heatmap of Numeric Predictors - ALL", line = 1, cex.main = 1.2)
```




```r
# Plot only strong correlations
corrplot(cor_matrix_90,
         method = "color",
         type = "lower",
         tl.col = "black",
```

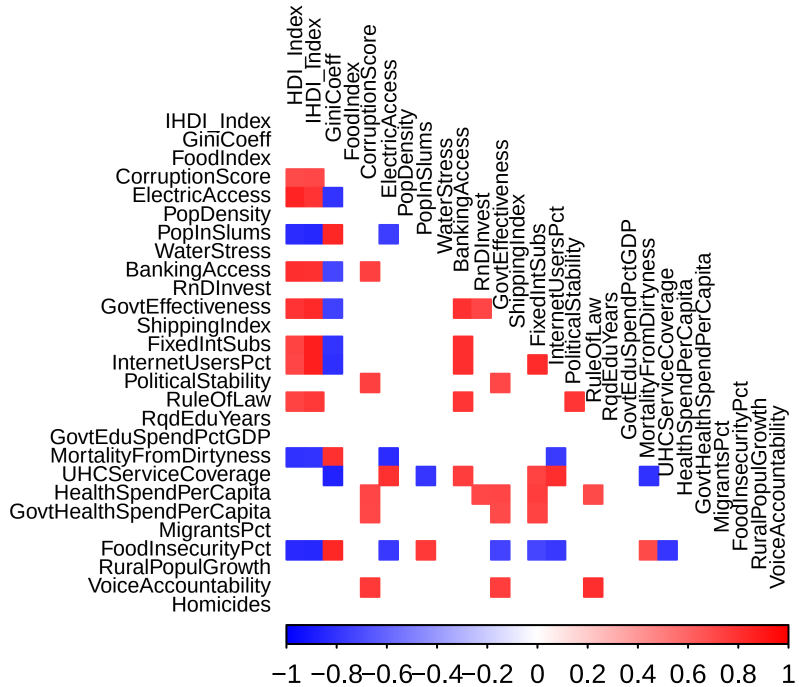
```r
        tl.cex = 0.7,
        col = colorRampPalette(c("blue", "white", "red"))(200),
        diag = FALSE,
        na.label = " ")  # blank for missing (non-displayed) values
title("Correlation Heatmap of Numeric Predictors - 90%+ correlation (DROP)", line = 1, cex.main = 1.2)
```




```r
corrplot(cor_matrix_70_90,
        method = "color",
        type = "lower",
        tl.col = "black",
        tl.cex = 0.7,
        col = colorRampPalette(c("blue", "white", "red"))(200),
        diag = FALSE,
        na.label = " ")  # blank for missing (non-displayed) values
```

```r
title("Correlation Heatmap of Numeric Predictors - 70%-90%+ correlation (VIF)", line = 1, cex.main = 1.2)
```

Correlation Heatmap of Numeric Predictors − 70%−90%+ correlation (V




```r
# Step 2: Compute Pearson correlation matrix using pairwise complete observations
cor_matrix <- cor(numeric_vars, use = "pairwise.complete.obs", method = "pearson")

# Step 3 Print
# Get total number of columns
cols <- ncol(cor_matrix)
col_names <- colnames(cor_matrix)

# Loop through and print in chunks
for (i in seq(1, cols, by = cols_per_table)) {
  # Subset the matrix
  end_col <- min(i + cols_per_table - 1, cols)
```

```
  sub_matrix <- cor_matrix[, i:end_col]

  # Print each chunk with a dynamic caption
  cat(
    asis_output(
      kable(
        round(sub_matrix, 2),
        format = "latex",
        booktabs = TRUE,
        caption = paste0("Correlation Matrix: Columns ", i, " to ", end_col)) %>%
        kable_styling(latex_options = c("HOLD_Position", "striped", "scale_down"))
      )
  )
}
```

```
# Step 1: Compute % missing per variable per year
missing_heatmap_data <- FirstCutData %>%
  group_by(year) %>%
  summarise(across(everything(), ~mean(is.na(.)) * 100)) %>%
  pivot_longer(-year, names_to = "variable", values_to = "pct_missing")

# Step 2: Plot with stoplight colors
stoplight <- c("#1a9641", "#ffea00", "#d7191c")
print(ggplot(missing_heatmap_data, aes(x = year, y = variable, fill = pct_missing)) +
  geom_tile(color = "white") +
  scale_fill_gradientn(
    colours = stoplight,
    values = scales::rescale(c(0, 25, 100)),
    limits = c(0, 100),
    name = "% Missing"
  ) +
  labs(
    title = "Missing Data Heatmap",
    x = "Year",
    y = "Variable"
  ) +
  theme_minimal(base_size = 11) +
  theme(
    axis.text.x = element_text(angle = 45, hjust = 1),
```

Table 8: Correlation Matrix: Columns 1 to 8

	HDI_Index	IHDI_Index	GiniCoeff	FoodIndex	CorruptionScore	ElectricAccess	PopDensity	PopInSlums
HDI_Index	1.00	0.98	-0.92	0.29	0.71	0.85	0.17	-0.82
IHDI_Index	0.98	1.00	-0.97	-0.10	0.72	0.80	0.14	-0.85
GiniCoeff	-0.92	-0.97	1.00	0.10	-0.62	-0.79	-0.09	0.84
FoodIndex	0.29	-0.10	0.10	1.00	0.13	0.20	0.08	-0.20
CorruptionScore	0.71	0.72	-0.62	0.13	1.00	0.47	0.20	-0.57
ElectricAccess	0.85	0.80	-0.79	0.20	0.47	1.00	0.10	-0.75
PopDensity	0.17	0.14	-0.09	0.08	0.20	0.10	1.00	-0.11
PopInSlums	-0.82	-0.85	0.84	-0.20	-0.57	-0.75	-0.11	1.00
WaterStress	0.12	0.08	-0.08	0.01	0.04	0.13	0.05	-0.14
BankingAccess	0.82	0.81	-0.72	0.04	0.75	0.56	0.18	-0.69
RnDInvest	0.63	0.62	-0.47	0.12	0.69	0.27	0.07	-0.47
GovtEffectiveness	0.79	0.83	-0.74	0.12	0.92	0.57	0.22	-0.64
ShippingIndex	0.46	0.46	-0.38	0.10	0.35	0.34	0.34	-0.36
FixedIntSubs	0.73	0.88	-0.80	0.14	0.64	0.47	0.24	-0.57
InternetUsersPct	0.72	0.87	-0.82	0.29	0.56	0.55	0.15	-0.64
PoliticalStability	0.59	0.63	-0.58	0.10	0.75	0.41	0.14	-0.56
RuleOfLaw	0.74	0.77	-0.68	0.13	0.94	0.50	0.18	-0.63
RqdEduYears	0.41	0.39	-0.38	0.13	0.26	0.42	-0.01	-0.36
GovtEduSpendPctGDP	0.26	0.28	-0.28	0.11	0.35	0.20	-0.15	-0.33
MortalityFromDirtyness	-0.80	-0.79	0.80	0.15	-0.42	-0.82	-0.07	1.00
UHCServiceCoverage	0.92	0.92	-0.86	0.25	0.61	0.81	0.13	-0.79
HealthSpendPerCapita	0.64	0.68	-0.57	0.09	0.73	0.36	0.26	-0.48
GovtHealthSpendPerCapita	0.63	0.67	-0.57	0.09	0.72	0.35	0.29	-0.47
MigrantsPct	0.41	0.50	-0.41	0.06	0.43	0.34	0.37	-0.38
FoodInsecurityPct	-0.83	-0.85	0.84	0.14	-0.62	-0.78	-0.13	0.77
RuralPopulGrowth	-0.24	-0.52	0.50	-0.02	-0.30	-0.23	-0.03	0.48
VoiceAccountability	0.59	0.65	-0.58	0.10	0.77	0.40	0.08	-0.55
Homicides	-0.24	-0.30	0.26	-0.01	-0.24	-0.06	-0.09	0.17

Table 9: Correlation Matrix: Columns 9 to 16

	WaterStress	BankingAccess	RnDInvest	GovtEffectiveness	ShippingIndex	FixedIntSubs	InternetUsersPct	PoliticalStability
HDI_Index	0.12	0.82	0.63	0.79	0.46	0.73	0.72	0.59
IHDI_Index	0.08	0.81	0.62	0.83	0.46	0.88	0.87	0.63
GiniCoeff	-0.08	-0.72	-0.47	-0.74	-0.38	-0.80	-0.82	-0.58
FoodIndex	0.01	0.04	0.12	0.12	0.10	0.14	0.29	0.10
CorruptionScore	0.04	0.75	0.69	0.92	0.35	0.64	0.56	0.75
ElectricAccess	0.13	0.56	0.27	0.57	0.34	0.47	0.55	0.41
PopDensity	0.05	0.18	0.07	0.22	0.34	0.24	0.15	0.14
PopInSlums	-0.14	-0.69	-0.47	-0.64	-0.36	-0.57	-0.64	-0.56
WaterStress	1.00	0.08	-0.08	0.04	0.05	-0.04	0.10	0.03
BankingAccess	0.08	1.00	0.66	0.80	0.49	0.81	0.81	0.65
RnDInvest	-0.08	0.66	1.00	0.72	0.46	0.59	0.49	0.43
GovtEffectiveness	0.04	0.80	0.72	1.00	0.49	0.69	0.62	0.71
ShippingIndex	0.05	0.49	0.46	0.49	1.00	0.49	0.46	0.09
FixedIntSubs	-0.04	0.81	0.59	0.69	0.49	1.00	0.83	0.46
InternetUsersPct	0.10	0.81	0.49	0.62	0.46	0.83	1.00	0.41
PoliticalStability	0.03	0.65	0.43	0.71	0.09	0.46	0.41	1.00
RuleOfLaw	0.05	0.78	0.70	0.94	0.39	0.65	0.57	0.78
RqdEduYears	-0.02	0.23	0.11	0.28	0.16	0.32	0.38	0.20
GovtEduSpendPctGDP	0.06	0.36	0.32	0.27	-0.02	0.12	0.17	0.33
MortalityFromDirtyness	-0.12	NA	-0.23	-0.57	-0.27	-0.56	-0.76	-0.45
UHCServiceCoverage	0.10	0.76	0.54	0.70	0.46	0.73	0.82	0.51
HealthSpendPerCapita	0.01	0.66	0.71	0.72	0.43	0.75	0.64	0.46
GovtHealthSpendPerCapita	0.02	0.66	0.69	0.71	0.39	0.74	0.63	0.46
MigrantsPct	0.61	NA	0.21	0.43	0.17	0.35	0.37	0.36
FoodInsecurityPct	-0.08	-0.69	-0.49	-0.74	-0.47	-0.72	-0.77	-0.51
RuralPopulGrowth	-0.13	-0.44	-0.05	-0.31	-0.24	-0.12	-0.16	-0.31
VoiceAccountability	-0.14	0.65	0.54	0.75	0.13	0.54	0.44	0.68
Homicides	-0.09	-0.28	-0.31	-0.24	-0.23	-0.22	-0.20	-0.18

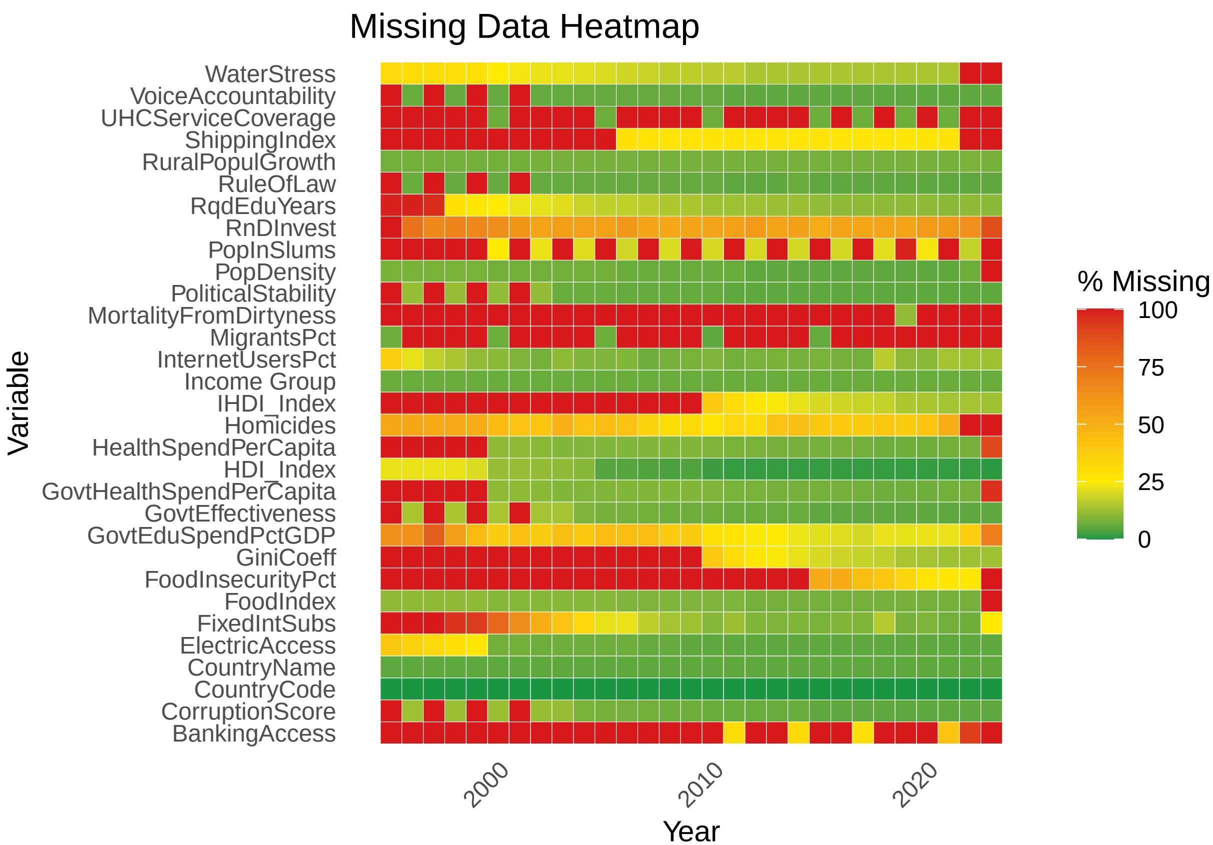
Table 10: Correlation Matrix: Columns 17 to 24

	RuleOfLaw	RqdEduYears	GovtEduSpendPctGDP	MortalityFromDirtyness	UHCServiceCoverage	HealthSpendPerCapita	GovtHealthSpendPerCapita	MigrantsPct
HDI_Index	0.74	0.41	0.26	-0.80	0.92	0.64	0.63	0.41
IHDI_Index	0.77	0.39	0.28	-0.79	0.92	0.68	0.67	0.50
GiniCoeff	-0.68	-0.38	-0.28	0.80	-0.86	-0.57	-0.57	-0.41
FoodIndex	0.13	0.13	0.11	0.15	0.25	0.09	0.09	0.06
CorruptionScore	0.94	0.26	0.35	-0.42	0.61	0.73	0.72	0.43
ElectricAccess	0.50	0.42	0.20	-0.82	0.81	0.36	0.35	0.34
PopDensity	0.18	-0.01	-0.15	-0.07	0.13	0.26	0.29	0.37
PopInSlums	-0.63	-0.36	-0.33	1.00	-0.79	-0.48	-0.47	-0.38
WaterStress	0.05	-0.02	0.06	-0.12	0.10	0.01	0.02	0.61
BankingAccess	0.78	0.23	0.36	NA	0.76	0.66	0.66	NA
RnDInvest	0.70	0.11	0.32	-0.23	0.54	0.71	0.69	0.21
GovtEffectiveness	0.94	0.28	0.27	-0.57	0.70	0.72	0.71	0.43
ShippingIndex	0.39	0.16	-0.02	-0.27	0.46	0.43	0.39	0.17
FixedIntSubs	0.65	0.32	0.12	-0.56	0.73	0.75	0.74	0.35
InternetUsersPct	0.57	0.38	0.17	-0.76	0.82	0.64	0.63	0.37
PoliticalStability	0.78	0.20	0.33	-0.45	0.51	0.46	0.46	0.36
RuleOfLaw	1.00	0.26	0.29	-0.47	0.62	0.71	0.70	0.42
RqdEduYears	0.26	1.00	0.16	-0.35	0.45	0.28	0.26	0.11
GovtEduSpendPctGDP	0.29	0.16	1.00	-0.13	0.25	0.20	0.21	-0.01
MortalityFromDirtyness	-0.47	-0.35	-0.13	1.00	-0.80	-0.35	-0.34	NA
UHCServiceCoverage	0.62	0.45	0.25	-0.80	1.00	0.58	0.56	0.35
HealthSpendPerCapita	0.71	0.28	0.20	-0.35	0.58	1.00	0.97	0.38
GovtHealthSpendPerCapita	0.70	0.26	0.21	-0.34	0.56	0.97	1.00	0.37
MigrantsPct	0.42	0.11	-0.01	NA	0.35	0.38	0.37	1.00
FoodInsecurityPct	-0.68	-0.30	-0.22	0.70	-0.79	-0.52	-0.53	-0.39
RuralPopulGrowth	-0.31	-0.12	-0.08	0.49	-0.46	-0.10	-0.11	-0.05
VoiceAccountability	0.82	0.32	0.31	-0.34	0.50	0.57	0.56	0.17
Homicides	-0.31	0.09	-0.04	0.11	-0.07	-0.24	-0.26	-0.21

Table 11: Correlation Matrix: Columns 25 to 28

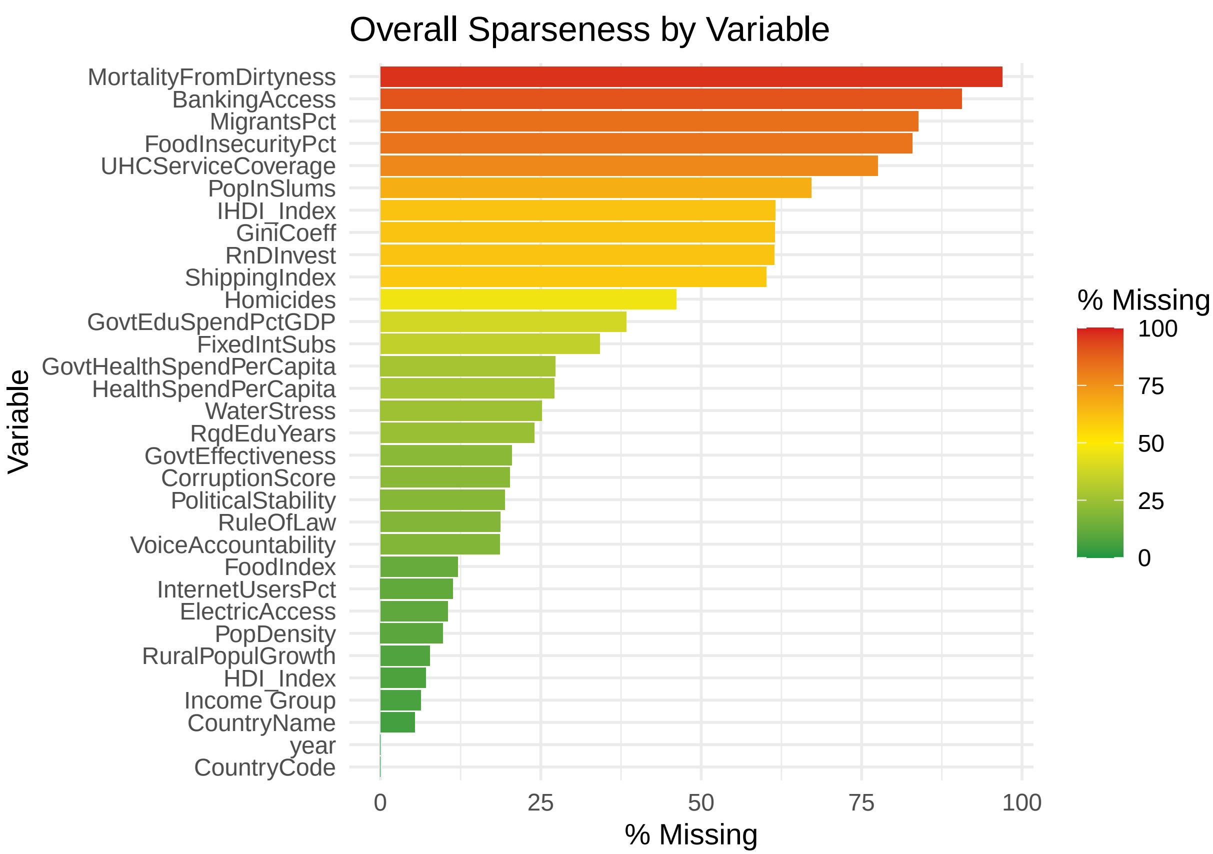
	FoodInsecurityPct	RuralPopulGrowth	VoiceAccountability	Homicides
HDI_Index	-0.83	-0.24	0.59	-0.24
IHDI_Index	-0.85	-0.52	0.65	-0.30
GiniCoeff	0.84	0.50	-0.58	0.26
FoodIndex	0.14	-0.02	0.10	-0.01
CorruptionScore	-0.62	-0.30	0.77	-0.24
ElectricAccess	-0.78	-0.23	0.40	-0.06
PopDensity	-0.13	-0.03	0.08	-0.09
PopInSlums	0.77	0.48	-0.55	0.17
WaterStress	-0.08	-0.13	-0.14	-0.09
BankingAccess	-0.69	-0.44	0.65	-0.28
RnDInvest	-0.49	-0.05	0.54	-0.31
GovtEffectiveness	-0.74	-0.31	0.75	-0.24
ShippingIndex	-0.47	-0.24	0.13	-0.23
FixedIntSubs	-0.72	-0.12	0.54	-0.22
InternetUsersPct	-0.77	-0.16	0.44	-0.20
PoliticalStability	-0.51	-0.31	0.68	-0.18
RuleOfLaw	-0.68	-0.31	0.82	-0.31
RqdEduYears	-0.30	-0.12	0.32	0.09
GovtEduSpendPctGDP	-0.22	-0.08	0.31	-0.04
MortalityFromDirtyness	0.70	0.49	-0.34	0.11
UHCServiceCoverage	-0.79	-0.46	0.50	-0.07
HealthSpendPerCapita	-0.52	-0.10	0.57	-0.24
GovtHealthSpendPerCapita	-0.53	-0.11	0.56	-0.26
MigrantsPct	-0.39	-0.05	0.17	-0.21
FoodInsecurityPct	1.00	0.43	-0.54	0.39
RuralPopulGrowth	0.43	1.00	-0.31	0.01
VoiceAccountability	-0.54	-0.31	1.00	-0.04
Homicides	0.39	0.01	-0.04	1.00

```
  panel.grid = element_blank()
))
```

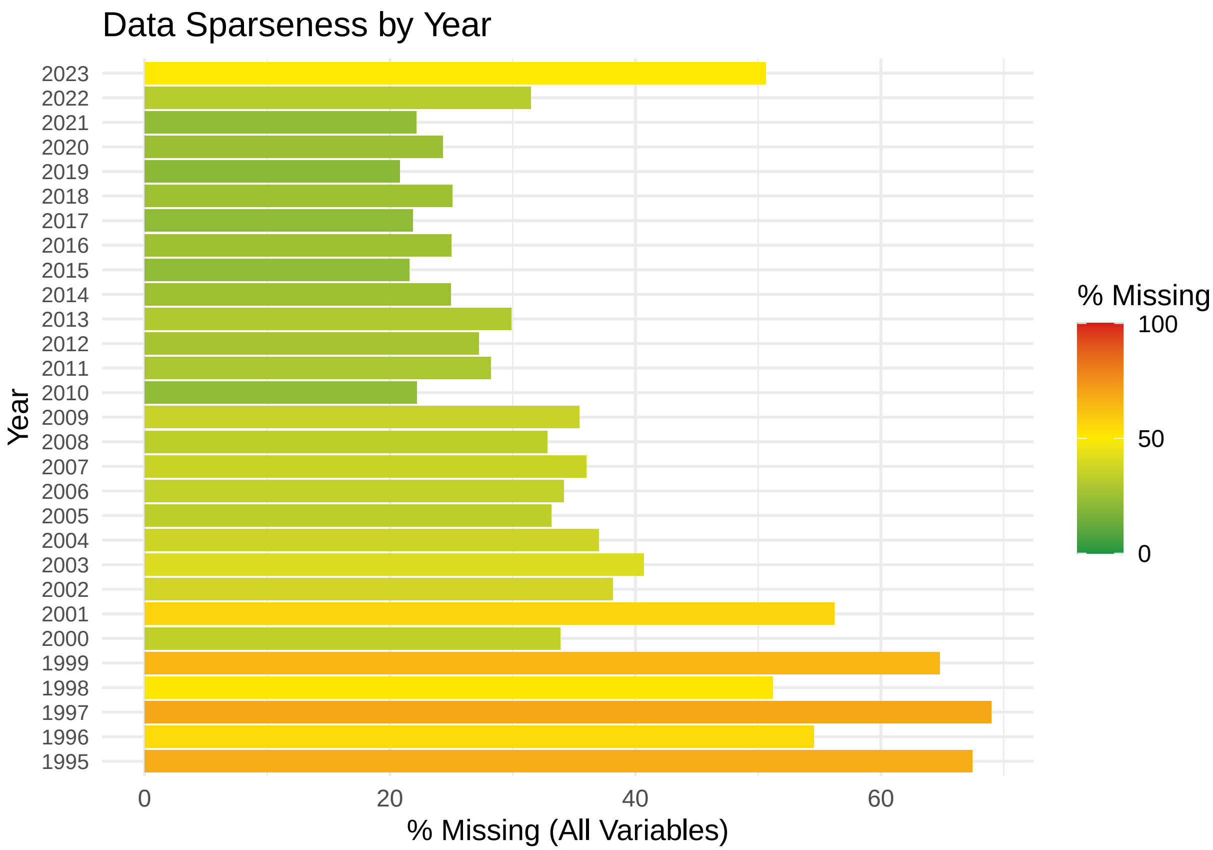



```
FirstCutData %>%
  summarise(across(everything(), ~ mean(is.na(.)) * 100)) %>%
  pivot_longer(cols = everything(), names_to = "Variable", values_to = "MissingPct") %>%
  ggplot(aes(x = reorder(Variable, MissingPct), y = MissingPct, fill = MissingPct)) +
  geom_col() +
  coord_flip() +
  scale_fill_gradientn(
    colors = stoplight,
    limits = c(0, 100),
```

```
    breaks = c(0, 25, 50, 75, 100),  # include 0 and 100
    labels = c("0", "25", "50", "75", "100"),
    name = "% Missing"
  ) +
  labs(
    title = "Overall Sparseness by Variable",
    x = "Variable",
    y = "% Missing"
  ) +
  theme_minimal()
```



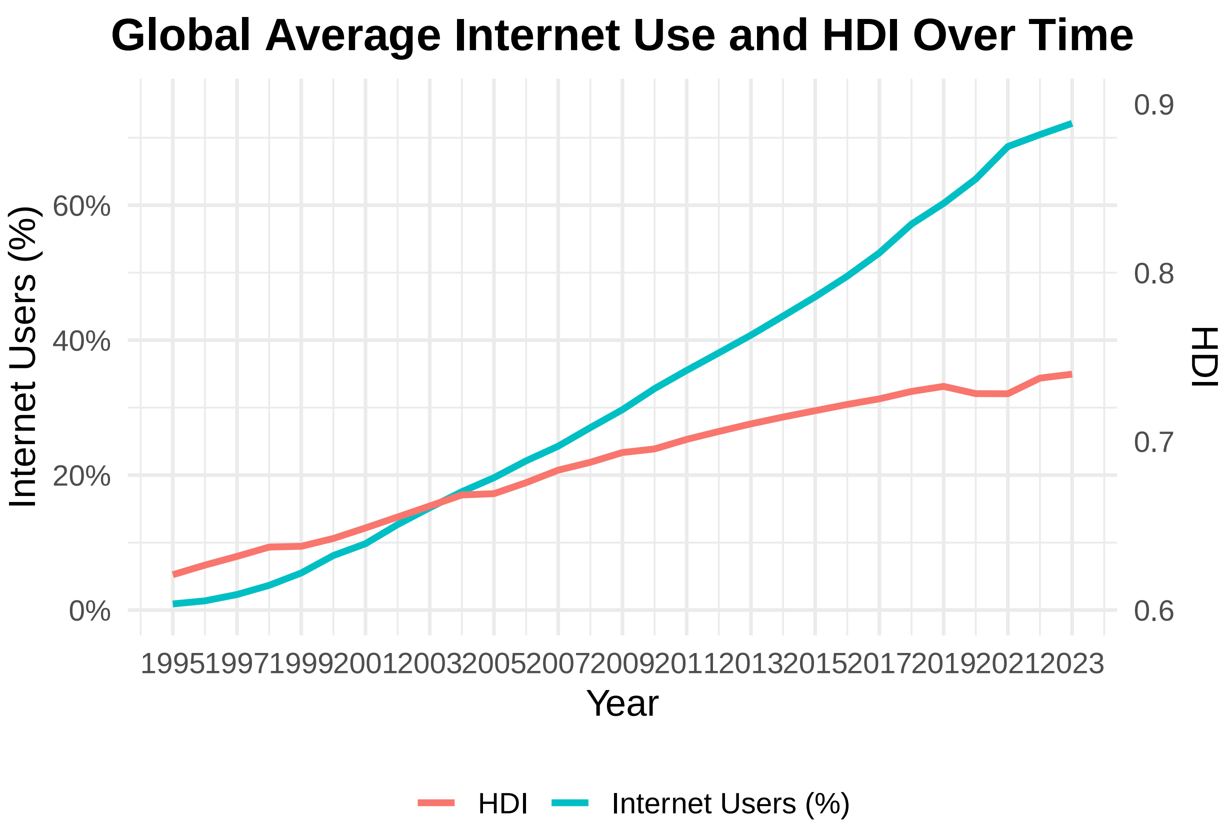
```
FirstCutData %>%
  group_by(year) %>%
  summarise(across(everything(), ~ mean(is.na(.)) * 100)) %>%
  pivot_longer(-year, names_to = "variable", values_to = "pct_missing") %>%
  group_by(year) %>%
  summarise(pct_missing = mean(pct_missing)) %>%
  arrange(year) %>%
  ggplot(aes(y = factor(year), x = pct_missing, fill = pct_missing)) +
  geom_col() +
  scale_fill_gradientn(
    colors = stoplight,
    values = rescale(c(0, 50, 100)),
    name = "% Missing",
    limits = c(0, 100),
    breaks = c(0, 50, 100),
    labels = c("0", "50", "100")
  ) +
  labs(
    y = "Year",
    x = "% Missing (All Variables)",
    title = "Data Sparseness by Year"
  ) +
  theme_minimal() +
  theme(
    axis.text.y = element_text(size = 8),
    legend.position = "right"
  )
```




```r
# Time series analysis

# Overall averages





# Prep data
FirstCut_Yearly_Avg <- FirstCutData %>%
  group_by(year) %>%
  summarize(
```

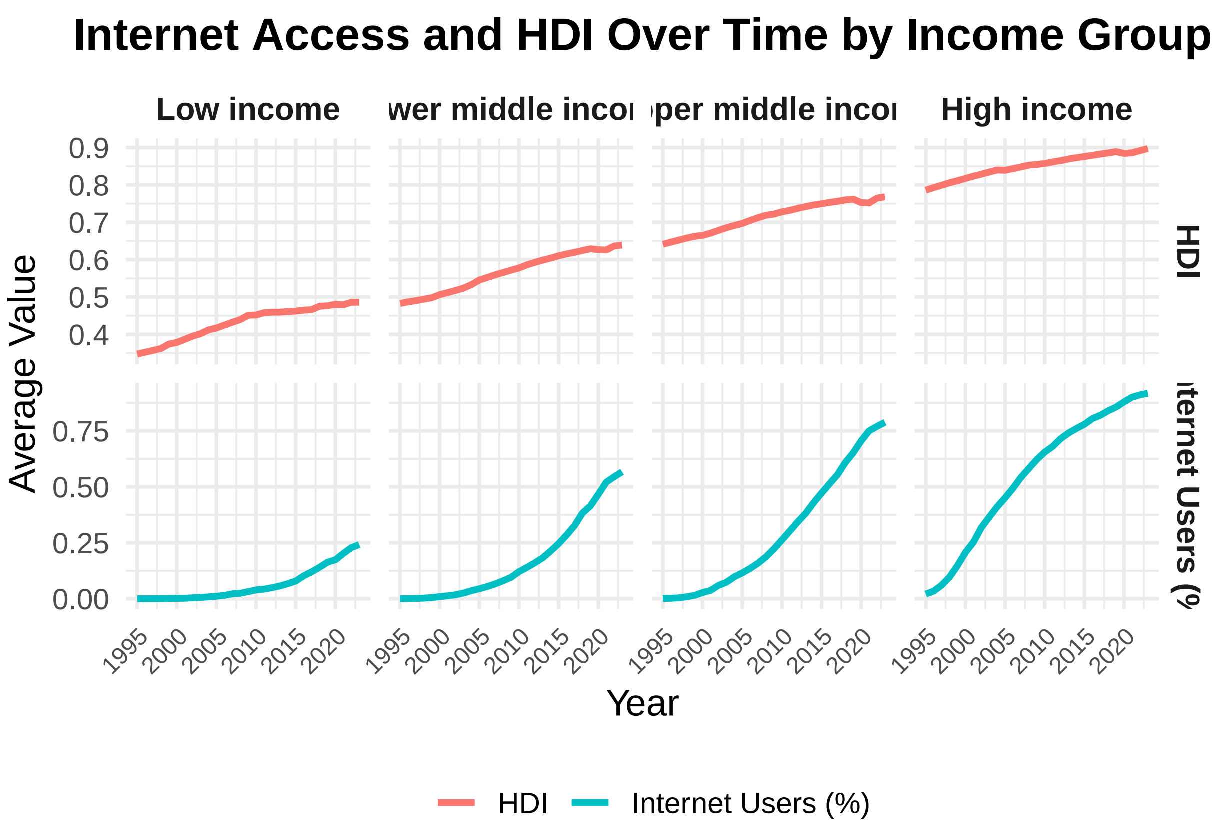
```r
    InternetUsers = mean(InternetUsersPct, na.rm = TRUE)/100,
    HDI = mean(HDI_Index, na.rm = TRUE),
    .groups = "drop"
  ) %>%
  mutate(
    HDI_scaled = (HDI - 0.6) / 0.4  # rescale onto same scale as inernet users
  )

# Plot with dual axes
ggplot(FirstCut_Yearly_Avg, aes(x = year)) +
  geom_line(aes(y = InternetUsers, color = "Internet Users (%)"), size = 1.2) +
  geom_line(aes(y = HDI_scaled, color = "HDI"), size = 1.2) +
  scale_x_continuous(breaks = seq(1995, 2025, by = 2))+
  scale_y_continuous(
    name = "Internet Users (%)",
    labels = scales::percent,
    limits = c(0, 0.75),  # force axis to run from 0% to 75%
    sec.axis = sec_axis(~ . * 0.4 + 0.6, name = "HDI") # label axis correctly
  ) +
  scale_color_manual(values = c("Internet Users (%)" = "#00BFC4", "HDI" = "#F8766D")) +
  labs(
    title = "Global Average Internet Use and HDI Over Time",
    x = "Year",
    color = ""
  ) +
  theme_minimal(base_size = 14) +
  theme(
    legend.position = "bottom",
    plot.title = element_text(face = "bold", hjust = 0.5)
  )
```

```
## Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use `linewidth` instead.
## This warning is displayed once every 8 hours.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```



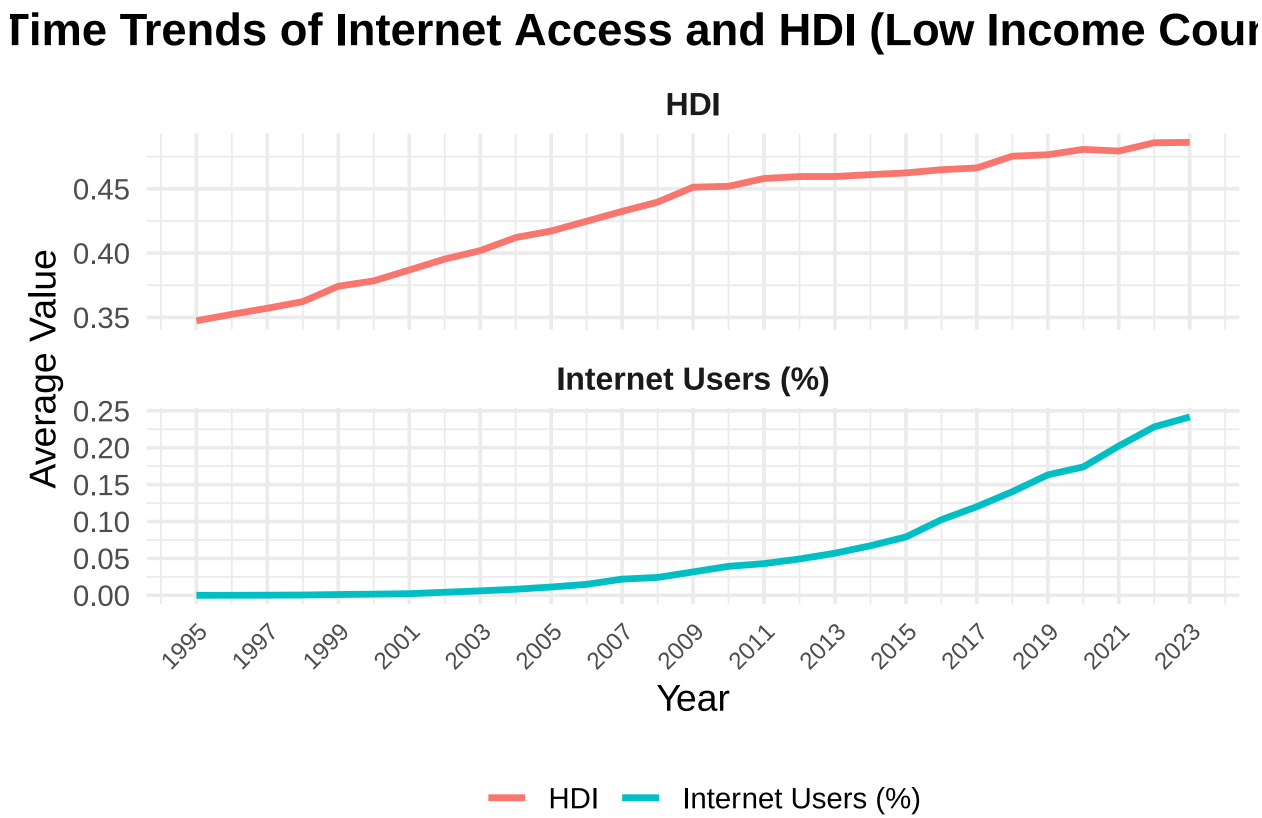

```
# Group and summarize
FirstCut_Yearly_Income_Avg <- FirstCutDataFiltered %>%
  group_by(year, `Income Group`) %>%
  summarize(
    `Internet Users (%)` = mean(InternetUsersPct, na.rm = TRUE)/100,
    `HDI` = mean(HDI_Index, na.rm = TRUE),
    .groups = "drop"
  )

# Pivot to long format
FirstCut_Yearly_Income_Avg_Long <- FirstCut_Yearly_Income_Avg %>%
  pivot_longer(cols = c(`Internet Users (%)`, `HDI`),
               names_to = "Metric", values_to = "Value")
```

```r
# Plot - facet by Income Group (columns) and Metric (rows), color by Metric
ggplot(FirstCut_Yearly_Income_Avg_Long,
       aes(x = year, y = Value, color = Metric)) +
  geom_line(size = 1.2) +
  facet_grid(Metric ~ `Income Group`, scales = "free_y") +
  scale_color_manual(values = c(
    "HDI" = "#F8766D",
    "Internet Users (%)" = "#00BFC4"
  )) +
  labs(
    title = "Internet Access and HDI Over Time by Income Group",
    x = "Year",
    y = "Average Value",
    color = ""
  ) +
  theme_minimal(base_size = 14) +
  theme(
    legend.position = "bottom",
    strip.text = element_text(face = "bold", size = 12),
    plot.title = element_text(face = "bold", hjust = 0.5),
    axis.text.x = element_text(size = 9, angle = 45, hjust = 1)  #   the fix
  )
```




```
# Filter to Low Income only
LowIncome_Yearly_Avg <- FirstCutDataFiltered %>%
  filter(`Income Group` == "Low income") %>%
  group_by(year) %>%
  summarize(
    `Internet Users (%)` = mean(InternetUsersPct, na.rm = TRUE)/100,
    `HDI` = mean(HDI_Index, na.rm = TRUE),
    .groups = "drop"
  ) %>%
  pivot_longer(cols = c(`Internet Users (%)`, `HDI`),
               names_to = "Metric", values_to = "Value")

# Plot with HDI and Internet Users (%) stacked vertically
```

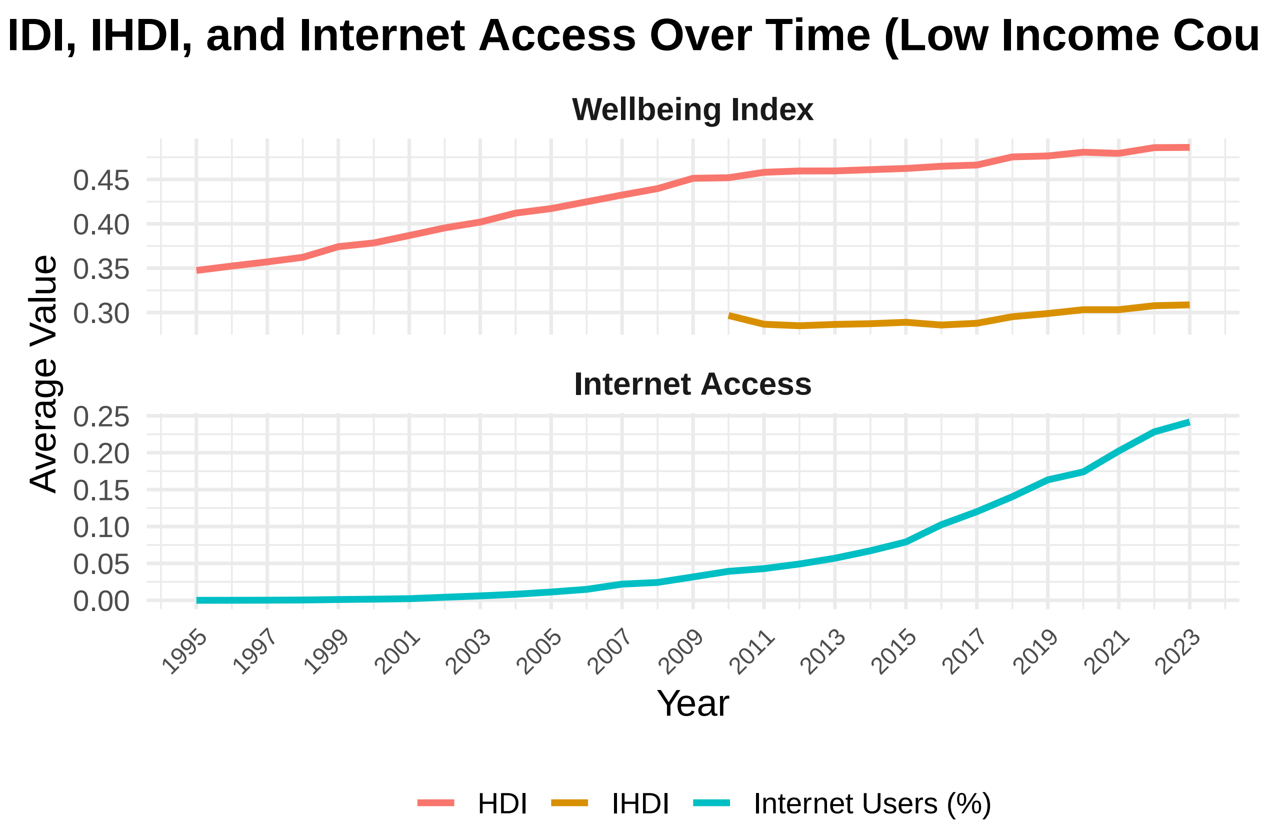
```r
ggplot(LowIncome_Yearly_Avg, aes(x = year, y = Value, color = Metric)) +
  geom_line(size = 1.2) +
  facet_wrap(~ Metric, ncol = 1, scales = "free_y") +
  scale_color_manual(values = c(
    "HDI" = "#F8766D",
    "Internet Users (%)" = "#00BFC4"
  )) +
  scale_x_continuous(breaks = seq(1995, 2025, 2)) +
  labs(
    title = "Time Trends of Internet Access and HDI (Low Income Countries)",
    x = "Year",
    y = "Average Value",
    color = ""
  ) +
  theme_minimal(base_size = 14) +
  theme(
    legend.position = "bottom",
    strip.text = element_text(face = "bold", size = 12),
    plot.title = element_text(face = "bold", hjust = 0.5),
    axis.text.x = element_text(size = 9, angle = 45, hjust = 1)
  )
```




```r
# Create groupings
LowIncome_Yearly_Avg <- FirstCutDataFiltered %>%
  filter(`Income Group` == "Low income") %>%
  group_by(year) %>%
  summarize(
    `Internet Users (%)` = mean(InternetUsersPct, na.rm = TRUE)/100,
    `HDI` = mean(HDI_Index, na.rm = TRUE),
    `IHDI` = mean(IHDI_Index, na.rm = TRUE),
    .groups = "drop"
  ) %>%
  pivot_longer(cols = c(`Internet Users (%)`, `HDI`, `IHDI`),
               names_to = "Metric", values_to = "Value") %>%
  mutate(
```

```r
    MetricGroup = ifelse(Metric %in% c("HDI", "IHDI"),
                         "Wellbeing Index",
                         "Internet Access")
  )

# Reorder facet levels so Wellbeing is on top
LowIncome_Yearly_Avg$MetricGroup <- factor(
  LowIncome_Yearly_Avg$MetricGroup,
  levels = c("Wellbeing Index", "Internet Access")
)

# Plot
ggplot(LowIncome_Yearly_Avg, aes(x = year, y = Value, color = Metric)) +
  geom_line(size = 1.2) +
  facet_wrap(~ MetricGroup, ncol = 1, scales = "free_y") +
  scale_color_manual(values = c(
    "HDI" = "#F8766D",
    "IHDI" = "#D89000",
    "Internet Users (%)" = "#00BFC4"
  )) +
  scale_x_continuous(breaks = seq(1995, 2025, 2)) +
  labs(
    title = "HDI, IHDI, and Internet Access Over Time (Low Income Countries)",
    x = "Year",
    y = "Average Value",
    color = ""
  ) +
  theme_minimal(base_size = 14) +
  theme(
    legend.position = "bottom",
    strip.text = element_text(face = "bold", size = 12),
    plot.title = element_text(face = "bold", hjust = 0.5),
    axis.text.x = element_text(size = 9, angle = 45, hjust = 1)
  )
```

```
## Warning: Removed 15 rows containing missing values or values outside the scale range
## (`geom_line()`).
```



IDI, IHDI, and Internet Access Over Time (Low Income Cou
Wellbeing Index
Internet Access
Average Value
Year
HDI
IHDI
Internet Users (%)